

Reliable Numerical Simulation of Financial SDEs

GBM Benchmarks and a Non-Affine Volatility Model

Team #<TEAM NUMBER>

Department of Applied Mathematics
Project 1: ODE IVP Extension Pathway, Financial SDE

18 September 2026

Project team

Source placeholders	Role
<Name 1>, <ID 1>	Project Manager
<Name 2>, <ID 2>	Mathematical Theory
<Name 3>, <ID 3>	Algorithm Implementation
<Name 4>, <ID 4>	Visualization & Report
<Name 5>, <ID 5>	Testing & Validation

Abstract

- ▶ GBM provides exact pathwise, moment and distribution benchmarks for EM and scalar Milstein.
- ▶ With 300,000 coupled paths, strong slopes are 0.4899 ± 0.0026 and 0.9099 ± 0.0167 . The analytic mean-bias slope is 0.9982 ± 0.0004 .
- ▶ At strong $\text{MAE} \leq 0.85$, EM uses 128 steps and Milstein uses 2 under the declared step-count measure.
- ▶ The non-affine model has empirical strong slope 0.5111 ± 0.0047 against a 2048-step numerical reference. Call-bias intervals include zero.

Contents

Abstract

1 Introduction and motivation

2 Financial SDE models

3 Discretisation and mathematical analysis

4 Implementation and reproducibility

5 Numerical experiments and results

6 Limitations and robustness

7 Conclusions

References

1.1 Problem background

- ▶ Numerical error must be separated from Monte Carlo uncertainty.
- ▶ GBM supplies exact transitions, moments and a terminal lognormal law [1–3].
- ▶ When no exact transition is available, nested Brownian grids measure agreement under refinement [4].
- ▶ A fine-grid solution remains a numerical reference. The report distinguishes theory, measurements and limitations.

1.2 Research questions and contribution

- ▶ Do coupled paths recover strong orders $1/2$ and 1 , and weak order 1 for the GBM mean?
- ▶ Which method needs fewer steps at a matched strong-error target?
- ▶ When do price-space updates lose positivity, and does the terminal law agree with the exact distribution?
- ▶ Does the non-affine model converge and pass increment, finiteness, positivity and reference checks?
- ▶ One auditable script fixes seeds, regenerates figures and numbers, and executes explicit PASS/FAIL checks.

1.3 Report map

- ▶ Section 2: financial models.
- ▶ Section 3: numerical schemes and convergence estimators.
- ▶ Section 4: implementation and validation design.
- ▶ Section 5: GBM and non-affine experiments.
- ▶ Sections 6–7: limitations and conclusions.
- ▶ Appendices A–E: reproduction, full source, AI log, contributions and evidence.

2.1 Geometric Brownian motion

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0. \quad (1)$$

For $f(s) = \log s$, Itô's formula gives

$$\begin{aligned} d \log S_t &= f'(S_t) dS_t + \frac{1}{2} f''(S_t) (dS_t)^2 \\ &= (\mu - \frac{1}{2} \sigma^2) dt + \sigma dW_t. \end{aligned} \quad (2)$$

W_t : standard Brownian motion. μ : physical-measure drift. σ : constant volatility.

2.1 Exact solution and moments

$$S_t = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right]. \quad (3)$$

$$\mathbb{E}[S_T] = S_0 e^{\mu T}, \quad (4)$$

$$\text{Var}(S_T) = S_0^2 e^{2\mu T} (e^{\sigma^2 T} - 1).$$

$$S_0 = 100, \quad \mu = 0.05, \quad \sigma = 0.40, \quad T = 1. \quad (5)$$

S_T is lognormal. The relatively high volatility exposes coarse-step positivity failures.

2.2 Non-affine stochastic volatility

Write $X_t = \log S_t$.

$$dX_t = (\mu - \tfrac{1}{2}g(Y_t)^2)dt + g(Y_t)dW_t^{(1)}, \quad (6)$$

$$dY_t = \kappa(\theta - Y_t)dt + \xi\sqrt{1 + Y_t^2} dW_t^{(2)}, \quad (7)$$

$$g(y) = \sigma_{\min} + \frac{\sigma_{\max} - \sigma_{\min}}{1 + e^{-y}}, \quad (8)$$

$$d\langle W^{(1)}, W^{(2)} \rangle_t = \rho dt.$$

2.2 Parameters and coefficient regularity

$$\begin{aligned} S_0 &= 100, & Y_0 &= 0, & \mu &= 0.05, & T &= 1, \\ \kappa &= 2, & \theta &= -0.2, & \xi &= 0.6, & \rho &= -0.7, \\ \sigma_{\min} &= 0.10, & \sigma_{\max} &= 0.50. \end{aligned} \tag{9}$$

- ▶ κ controls mean reversion of Y toward θ . ξ controls its diffusion strength. ρ sets the noise correlation.
- ▶ The logistic map gives $0.10 < g(y) < 0.50$ and a bounded derivative.
- ▶ The derivative of $\sqrt{1+y^2}$ has magnitude at most one. The coefficients meet the regularity and growth structure used in standard EM arguments.
- ▶ The finite reference below is numerical, not a closed-form oracle.

3.1 Euler–Maruyama for GBM

Let $t_n = nh$, $h = T/N$, and $\Delta W_n \sim N(0, h)$. Freeze both coefficients at the left endpoint.

$$\begin{aligned} S_{n+1}^{EM} &= S_n^{EM} + \mu S_n^{EM} h + \sigma S_n^{EM} \Delta W_n \\ &= S_n^{EM} (1 + \mu h + \sigma \Delta W_n). \end{aligned} \tag{10}$$

- ▶ Globally Lipschitz drift and diffusion give strong order $1/2$.
- ▶ Weak order 1 holds for sufficiently smooth test functions [1,2].
- ▶ The GBM coefficients are globally Lipschitz.

3.2 Milstein for GBM

The scalar Itô–Taylor expansion retains

$$\int_{t_n}^{t_{n+1}} \int_{t_n}^s dW_r dW_s = \frac{1}{2}[(\Delta W_n)^2 - h]. \quad (11)$$

For $b(s) = \sigma s$ and $b'(s) = \sigma$,

$$\begin{aligned} S_{n+1}^{Mil} = & S_n^{Mil} + \mu S_n^{Mil} h + \sigma S_n^{Mil} \Delta W_n \\ & + \frac{1}{2} \sigma^2 S_n^{Mil} [(\Delta W_n)^2 - h]. \end{aligned} \quad (12)$$

The extra term raises scalar strong order to 1, while weak order remains 1 [3].

3.3 Exact transition and pathwise coupling

$$S_{n+1}^{exact} = S_n^{exact} \exp[(\mu - \frac{1}{2}\sigma^2)h + \sigma\Delta W_n]. \quad (13)$$

Exact, EM and Milstein paths share every Brownian increment.

$$\hat{e}_s(h) = \frac{1}{M} \sum_{m=1}^M \left| S_{N,m}^{(h)} - S_{T,m}^{exact} \right|. \quad (14)$$

- ▶ A log-log least-squares slope estimates p in $\mathbb{E}|S_N - S_T| = O(h^p)$.
- ▶ Common random drivers isolate discretisation error. Independent exact paths do not provide this pathwise comparison.

3.4 Weak error and an analytic bias oracle

$$e_w(h) = \left| \mathbb{E}\phi(S_N^{(h)}) - \mathbb{E}\phi(S_T) \right|. \quad (15)$$

For $\phi(s) = s$, the EM and Milstein multipliers both have conditional mean $1 + \mu h$.

$$\mathbb{E}[S_N^{EM}] = \mathbb{E}[S_N^{Mil}] = S_0(1 + \mu h)^N. \quad (16)$$

This gives a deterministic mean-bias oracle independent of Monte Carlo noise.

3.4 First-order mean bias

Use $N = T/h$ and $\log(1 + \mu h) = \mu h - \frac{1}{2}\mu^2 h^2 + O(h^3)$.

$$S_0(1 + \mu h)^{T/h} = S_0 \exp\left(\mu T - \frac{1}{2}\mu^2 Th + O(h^2)\right). \quad (17)$$

$$e_w(h) = \frac{1}{2}S_0 e^{\mu T} \mu^2 Th + O(h^2). \quad (18)$$

- For the benchmark parameters, the mean bias has asymptotic order one.
- The code also reports a paired Monte Carlo confidence interval for $S_N - S_T$.
- Analytic bias prevents sampling noise from hiding the weak rate.

3.5 Positivity analysis: EM

An EM step is positive only if

$$1 + \mu h + \sigma \sqrt{h} Z_n > 0. \quad (19)$$

$$p_{neg,EM}(h) = \Phi\left(-\frac{1 + \mu h}{\sigma \sqrt{h}}\right) > 0. \quad (20)$$

- ▶ Φ is the standard normal cumulative distribution function.
- ▶ Gaussian tails allow a negative multiplier even when the true GBM stays positive.
- ▶ No observed violation in a finite sample is not a positivity guarantee.

3.5 Positivity analysis: Milstein

$$q(z) = \frac{1}{2}\sigma^2 h z^2 + \sigma\sqrt{h}z + 1 + \mu h - \frac{1}{2}\sigma^2 h. \quad (21)$$

$$q_{\min} = \frac{1}{2} + \mu h - \frac{1}{2}\sigma^2 h. \quad (22)$$

- For the benchmark and $0 < h \leq 1$, this minimum is positive.
- The result depends on parameters and step size. Milstein does not preserve positivity unconditionally.
- Exact exponential and log-price simulation preserve positivity mathematically.

3.6 Correlated EM: Brownian increments

Generate independent $Z_{1,n}, Z_{2,n} \sim N(0, 1)$ and set

$$\Delta W_n^{(1)} = \sqrt{h} Z_{1,n}, \quad (23)$$

$$\Delta W_n^{(2)} = \sqrt{h} \left(\rho Z_{1,n} + \sqrt{1 - \rho^2} Z_{2,n} \right). \quad (24)$$

Both increments have variance h , and their correlation equals ρ .

3.6 Correlated EM: state updates

$$X_{n+1} = X_n + (\mu - \frac{1}{2}g(Y_n)^2)h + g(Y_n)\Delta W_n^{(1)}, \quad (25)$$

$$Y_{n+1} = Y_n + \kappa(\theta - Y_n)h + \xi\sqrt{1 + Y_n^2}\Delta W_n^{(2)}, \quad (26)$$

$$S_{n+1} = e^{X_{n+1}} > 0. \quad (27)$$

Two algebraically equivalent branches evaluate the logistic function without overflow for large $|Y_n|$.

3.6 Why scalar Milstein is not applied

- ▶ The non-affine model has two stochastic factors.
- ▶ In independent Brownian drivers, the diffusion vector fields have cross derivatives through $g(Y)$ and $\sqrt{1 + Y^2}$.
- ▶ A full multidimensional Milstein method generally needs cross terms and simulated iterated stochastic integrals.
- ▶ Omitting these terms cannot justify a strong-order-one claim.

3.7 Nested-grid numerical reference

Let $h_{ref} = T/N_{ref}$ and $h = Kh_{ref}$.

$$\Delta W_{n,coarse}^{(i)} = \sum_{j=0}^{K-1} \Delta W_{Kn+j,ref}^{(i)}, \quad i = 1, 2. \quad (28)$$

- Coarse and fine solutions represent the same Brownian path.
- Main reference: $N_{ref} = 2048$. Independent refinement: 4096 steps on 4000 fresh paths.
- Paired differences for $P(S_T) = (S_T - 100)^+$ reduce variance. A weak trend is reported only if its confidence interval is separated from zero.

3.8 Uncertainty quantification

For independent path-level observations $D_1, \dots, D_M$,

$$\bar{D} \pm 1.96 \frac{s_D}{\sqrt{M}}, \quad s_D^2 = \frac{1}{M-1} \sum_{m=1}^M (D_m - \bar{D})^2. \quad (29)$$

- ▶ Monte Carlo intervals quantify uncertainty in path-level sample means.
- ▶ Bands around fitted log-log lines are regression diagnostics.
- ▶ Slope standard errors are conditional on the chosen mesh range, not repeated Monte Carlo uncertainty.

4.1 Core algorithm

Algorithm 1. Coupled non-affine EM experiment

- ▶ Input: parameters, seed, M , N_{ref} and coarse step counts.
- ▶ For each memory-safe batch, draw fine-grid independent normal arrays and construct correlated increments.
- ▶ Integrate the reference path. For each coarse count, aggregate increments and integrate the coarse path.
- ▶ Store paired terminal differences and combine batches.
- ▶ Estimate means and confidence intervals. Validate increment variance/correlation, finiteness and positivity.
- ▶ GBM uses the same coupling pattern with an exact transition.

4.2 Single-file code structure

Function group	Purpose
<code>gbm_terminal_coupled</code> , <code>run_gbm</code>	Exact/EM/Milstein coupling, errors, positivity and matched cost.
<code>logistic_vol</code> , <code>em_nonaffine_from_increments</code>	Stable coefficients and integration from supplied increments.
<code>aggregate_increments</code> , <code>run_nonaffine</code>	Nested coupling, reference comparison, payoff and diagnostics.
<code>nonaffine_reference_refinement</code>	Independent 2048-to-4096 check.
<code>plot_*</code> , <code>validate</code>	Five figures and ten PASS/FAIL checks.
Integrators receive increments explicitly, separating random drivers from numerical methods.	

4.3 Reproducible protocol

- ▶ Seed: 20260918. GBM: $M = 300,000$. Non-affine: $M = 20,000$.
- ▶ Batches contain at most 1000 paths. $N_{ref} = 2048$.
- ▶ Run `python financial_sde_project.py`.
- ▶ Outputs: `latex/generated/results.json`, `run_output.txt`, and Figures 1–5 at 220 DPI.
- ▶ All paths are relative. `-quick` is a reduced smoke test only.
- ▶ NumPy vectorises path operations [5]. Matplotlib creates figures [6].

4.4 Oracle chain and automatic checks

- ▶ Independent GBM oracles: exact transition and closed-form terminal moments/distribution.
- ▶ An independent finer grid checks the non-affine reference.
- ▶ Ten checks cover order bands, the exact GBM mean, two increment variances, correlation, finiteness, positivity and a plausible non-affine strong slope.
- ▶ All ten pass in the archived output. Acceptance bands allow sampling variation while detecting wrong scaling, correlation or schemes.

4.5 Declared cost measure

- ▶ Cost equals time steps per path, or one drift/diffusion coefficient evaluation per step in the scalar GBM loop.
- ▶ Excluded: plotting, random-number setup, file I/O, compilation and vector-allocation overhead.
- ▶ Milstein evaluates an additional correction, so the two step types do not have equal floating-point costs.
- ▶ The comparison concerns step counts, not measured wall-clock speed.

5.1 GBM strong and weak convergence

- ▶ 300,000 coupled paths. EM strong slope: 0.4899 ± 0.0026 .
- ▶ Milstein strong slope: 0.9099 ± 0.0167 .
- ▶ The Milstein shortfall from one is consistent with a finite pre-asymptotic mesh range.
- ▶ Analytic mean-bias slope: 0.9982 ± 0.0004 .
- ▶ Figure 1 shows reference slopes and 95% fit bands. The reported $\pm$ values are slope standard errors.

5.1 Table 2: GBM convergence

N	h	EM MAE	Milstein MAE	Mean bias
4	.250000	4.52887	.460308	.0325759
8	.125000	3.28870	.268833	.0163567
16	.062500	2.34715	.149881	.00819567
32	.031250	1.67424	.0800666	.00410218
64	.015625	1.18565	.0414826	.00205218
128	.0078125	.838329	.0211342	.00102636
256	.00390625	.594648	.0106906	.000513248

The analytic weak-bias column is common to EM and Milstein for $\phi(s) = s$. All errors are in price units.

5.1 Figure 1: coupled GBM convergence



assets/figure1.pdf

Terminal MAE follows the $h^{1/2}$ and h reference trends. Analytic mean bias follows order one, without Monte Carlo noise.

5.2 Matched-accuracy cost comparison

Predeclared target: $\mathbb{E}|S_N - S_T| \leq 0.85$.

- ▶ EM: the coarsest passing tested grid has $N = 128$, MAE 0.83833.
- ▶ Milstein: $N = 2$, MAE 0.75433.

$$\frac{C_{EM}}{C_{Mil}} = \frac{128}{2} = 64. \quad (30)$$

This is a step-count ratio. At much looser accuracy, EM's cheaper algebra could matter, but that regime is not measured here.

5.3 Terminal distribution and moment validation

$$\mathbb{E}[S_T] = 105.127110, \quad \text{Var}(S_T) = 1917.591686. \quad (31)$$

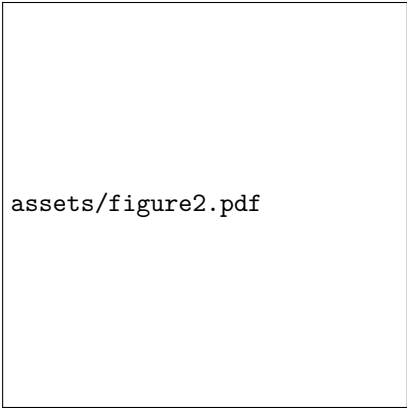
- ▶ Exact-transition samples give mean 105.088894 and variance 1915.728882.
- ▶ The sample mean lies within four Monte Carlo standard errors of the formula.
- ▶ At $N = 64$, Table 3 compares the 1%, 5%, 50%, 95% and 99% terminal quantiles.
- ▶ Density and quantile plots check the full distribution and remaining tail discrepancies.

5.3 Table 3: terminal quantiles

Method	1%	5%	50%	95%	99%
Exact lognormal	38.2687	50.2610	97.0446	187.3749	246.0930
EM, $N = 64$	37.9146	50.0592	97.0464	187.5114	244.2812
Milstein, $N = 64$	38.3043	50.2674	96.9520	187.6476	245.1338

Values are in price units. Central quantiles agree closely, while the tails expose finite-step error more sensitively.

5.3 Figure 2: terminal-law validation



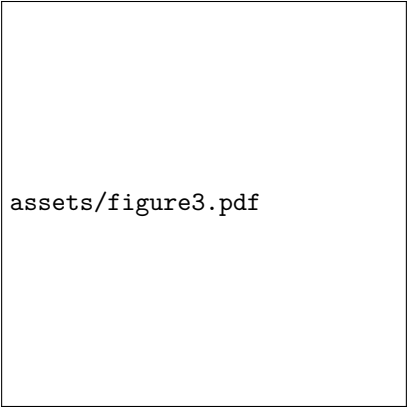
assets/figure2.pdf

Density overlays check the distribution beyond its first two moments. Quantile comparisons expose residual finite-step and sampling differences.

5.4 Positivity stress test

- ▶ At $h = 1$, 0.435% of EM paths encounter a non-positive multiplier.
- ▶ At $h = 1/2$, the observed fraction is 0.034%.
- ▶ For $h \leq 1/4$, no violation occurs among 300,000 paths.
- ▶ No Milstein violation occurs for the tested parameters, consistent with $q_{\min} > 0$.
- ▶ Zero observed failures do not imply zero failure probability for EM.

5.4 Figure 3: non-positive multipliers



assets/figure3.pdf

Fraction of paths with at least one non-positive update. Milstein is positive here because the quadratic minimum is positive on the tested range.

5.5 Non-affine nested-grid convergence

- ▶ 20,000 paths compare coarse grids with the coupled $N_{ref} = 2048$ reference.
- ▶ Fitting the first four rows gives strong slope 0.5111 ± 0.0047 .
- ▶ The final row is a near-reference diagnostic and is excluded from the slope fit.
- ▶ Every 95% interval for the paired call-payoff difference includes zero.
- ▶ Sampling uncertainty is comparable to weak bias. These data do not support a quoted weak slope.

5.5 Table 4: strong errors

N	h	Strong MAE	95% CI
16	.062500	.886321	[.875404, .897238]
32	.031250	.627466	[.619890, .635042]
64	.015625	.441095	[.435856, .446335]
128	.0078125	.306001	[.302342, .309660]
256	.00390625	.210677	[.208183, .213171]

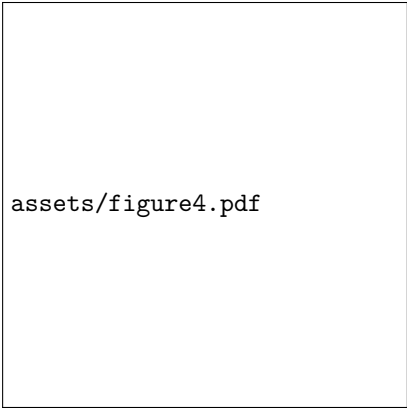
All errors are relative to the numerical reference and expressed in price units.

5.5 Table 4: paired call-payoff bias

N	Call bias	95% CI
16	.004204	$[-.010095, .018504]$
32	.003974	$[-.006056, .014005]$
64	.003689	$[-.003286, .010664]$
128	.002141	$[-.002714, .006996]$
256	.002748	$[-.000568, .006064]$

Payoff $P(S_T) = (S_T - 100)^+$. The bias is not statistically distinguishable from zero at this Monte Carlo precision.

5.5 Figure 4: nested-grid evidence



assets/figure4.pdf

Pathwise terminal error follows an empirical half-order trend. All payoff-bias intervals cross zero, preventing a supported weak-order estimate.

5.6 Increment, finiteness and positivity checks

$$\begin{aligned}\frac{\widehat{\text{Var}}(\Delta W^{(1)})}{h} &= 1.00526, & \frac{\widehat{\text{Var}}(\Delta W^{(2)})}{h} &= 1.00346, \\ \widehat{\text{Corr}} &= -0.70185.\end{aligned}\tag{32}$$

- ▶ Targets are $(1, 1, -0.7)$. All 20,000 reference paths are finite and prices are positive.
- ▶ Mean terminal price: 105.01815, standard error 0.21433.
- ▶ Mean call payoff: 14.48371, standard error 0.15090.
- ▶ These are model outputs, not closed-form benchmarks.

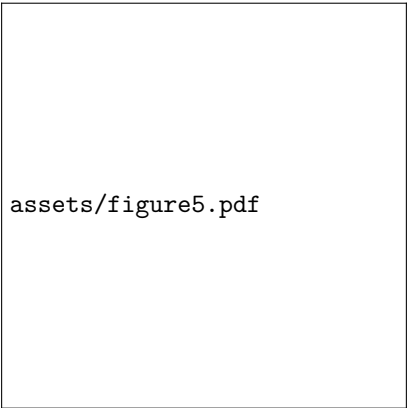
5.6 Reference-refinement check

On 4000 fresh coupled paths, compare 2048 and 4096 steps.

$$\begin{aligned}\text{Mean absolute terminal difference} &= 0.057626, \\ 95\% \text{ CI} &= [0.056114, 0.059139].\end{aligned}\tag{33}$$

- ▶ The change is much smaller than the coarsest errors, but about 27% of the $N = 256$ error.
- ▶ The reference supports the observed trend but remains inexact.
- ▶ A finer reference is needed before reporting more terminal digits or resolving weak payoff bias.

5.6 Figure 5: non-affine paths



assets/figure5.pdf

Exponentiating X yields positive prices. The logistic map keeps volatility between 0.10 and 0.50 $\text{year}^{-1/2}$.

6 Limitations and robustness (1/2)

- ▶ Numerical reference: 2048-step EM has residual bias. Refinement quantifies differences without eliminating that bias.
- ▶ Weak convergence: paired samples still leave call-bias intervals across zero. More paths, multilevel Monte Carlo or another payoff may be needed.
- ▶ Regression: slope standard errors depend on the mesh range. They are neither theorem proofs nor complete model-selection uncertainty.

6 Limitations and robustness (2/2)

- ▶ Cost: step counts omit extra Milstein arithmetic and hardware effects. Wall-clock comparisons require repeated same-machine timings and separate operation counts.
- ▶ Model risk: numerical convergence does not establish market fit. Calibration, risk-neutral drift, costs, jumps and empirical validation are outside scope.
- ▶ Extreme paths: stable logistic evaluation and finiteness checks help, but e^x can overflow for extreme parameters or horizons. Production code needs an overflow policy.

7 Conclusions

- ▶ GBM strong slopes: EM 0.4899 ± 0.0026 , Milstein 0.9099 ± 0.0167 . Analytic mean weak slope: 0.9982 ± 0.0004 .
- ▶ At matched MAE 0.85, Milstein uses 2 steps and EM 128. This establishes a 64:1 step-count ratio.
- ▶ Distribution and moment checks agree with the exact lognormal benchmark. Coarse-grid EM can lose positivity.
- ▶ Non-affine nested coupling gives empirical half-order strong convergence and passes diagnostics. Weak payoff bias remains unresolved.

7 Extensions

- ▶ Multilevel Monte Carlo can reuse nested coupling to reduce the cost of resolving weak payoffs [4].
- ▶ A correctly implemented multidimensional Milstein method would include the required iterated stochastic integrals.

References (1/2)

1. D. J. Higham. An algorithmic introduction to numerical simulation of stochastic differential equations. *SIAM Review* 43(3), 525–546, 2001. DOI: 10.1137/S0036144500378302.
2. P. E. Kloeden and E. Platen. *Numerical Solution of Stochastic Differential Equations*. Springer, 1992. DOI: 10.1007/978-3-662-12616-5.
3. G. N. Milstein. *Numerical Integration of Stochastic Differential Equations*. Springer, 1995. DOI: 10.1007/978-94-015-8455-5.
4. M. B. Giles. Multilevel Monte Carlo path simulation. *Operations Research* 56(3), 607–617, 2008. DOI: 10.1287/opre.1070.0496.

References (2/2)

5. C. R. Harris et al. Array programming with NumPy. *Nature* 585, 357–362, 2020. DOI: 10.1038/s41586-020-2649-2.
6. J. D. Hunter. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering* 9(3), 90–95, 2007. DOI: 10.1109/MCSE.2007.55.
7. MTH321 teaching team. *Project 1 Brief: Numerical Solution of ODE Initial Value Problems*, extension pathway 4, 2026. Course material supplied with the project.
8. MTH321 teaching team. *Problem Pack: Five Suggested Directions for the ODE IVP Project*, Financial SDE: GBM benchmark and non-affine volatility, 2026. Course material supplied with the project.

A.1 Environment and commands

Install dependencies in an isolated Python environment:

```
python -m pip install numpy matplotlib
```

From the original repository root:

```
python financial_sde_project.py
```

The output directory is created automatically. JSON records versions, platform, seed, sample sizes, parameters, table rows, validation and wall time.

A.2 Archived program output (1/2)

A.2 Archived program output (2/2)

B Complete Python source code: lines on report p.14

B Source code: report p.15 (1/3)

B Source code: report p.15 (2/3)

B Source code: report p.15 (3/3)

B Source code: report p.16 (1/3)

B Source code: report p.16 (2/3)

B Source code: report p.16 (3/3)

B Source code: report p.17 (1/3)

B Source code: report p.17 (2/3)

B Source code: report p.17 (3/3)

B Source code: report p.18 (1/3)

B Source code: report p.18 (2/3)

B Source code: report p.18 (3/3)

B Source code: report p.19 (1/3)

B Source code: report p.19 (2/3)

B Source code: report p.19 (3/3)

B Source code: report p.20 (1/3)

B Source code: report p.20 (2/3)

B Source code: report p.20 (3/3)

B Source code: report p.21 (1/3)

B Source code: report p.21 (2/3)

B Source code: report p.21 (3/3)

B Source code: report p.22 (1/3)

B Source code: report p.22 (2/3)

B Source code: report p.22 (3/3)

B Source code: report p.23 (1/3)

B Source code: report p.23 (2/3)

B Source code: report p.23 (3/3)

C.1 Tools used

C.2 Significant AI-assisted claim audits

C.3 Independent verification and decisions

C.4 Intellectual ownership declaration

C.4 Intellectual ownership: signature table

D Individual contribution: project manager

D Individual contribution: Mathematical theory

D Individual contribution: Algorithm implementation

D Individual contribution: Visualization and report

D Individual contribution: Testing and validation

E Submission evidence matrix